

Joint Cascade Vulnerability Assessment of Interdependent Power–Water Infrastructures

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Abstract—The joint operation and coordination of power and water infrastructures have made them more vulnerable than before to cyber-physical attacks. This is because each system relies on the other to provide its services. Such a mutual dependence increases the risk of large-scale cascading failures. Therefore, it becomes of utmost importance to analyze the vulnerability of interdependent power–water infrastructures. To accurately quantify the overall damage due to cascading failure propagation, the vulnerability assessment should consider not only the topological features but also the power dynamics and the hydraulic features. Existing works fail to identify the combination of nodes that present the greatest damage. To fill up this gap, this article proposes and examines a low-complexity vulnerability assessment strategy based on the most critical nodes, which maximizes the overall damage. Experiments are conducted on two different power–water systems. It is shown that the proposed strategy identifies the most vulnerable nodes that cause a high damage impact from either or both systems, but with a much lower computational time compared to the one that selects the target nodes based on how much damage they inflict upon the system. A visual assessment of the physical disturbances under the most vulnerable nodes is presented.

Index Terms—Cascading failure, interdependent infrastructures, power grid, vulnerability, water distribution system (WDS).

NOMENCLATURE

Acronyms

AC	Alternating current.
AHP	Analytical Hierarchical process.
CFS	Cascading failure simulator.
DMAs	District metered areas.
DSR	Demand satisfaction ratio.
FDIA	False data injection attack.
MCN	Most critical nodes.
PDF	Probability density function.
PLCs	Programmable logic controllers.
PMUs	Phasor measurement units.

Manuscript received 13 November 2022; revised 14 March 2023 and 4 August 2023; accepted 7 September 2023. Date of publication 22 September 2023; date of current version 12 December 2023. This work was supported by the National Science Foundation Energy, Power, Control, and Networks Program under Grant 2220346 and Grant 2220347. (Corresponding author: Rachad Atat.)

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Digital Object Identifier 10.1109/JSYST.2023.3314099

PN	Power node.
PoDN	Percentage of drop in net-ability.
SCADA	Supervisory control and data acquisition.
WDS	Water distribution system.
WN	Water node.
<i>Parameters</i>	
η_P, η_W	Number of metrics for the power and water systems, respectively.
$\mathbf{x}, \mathbf{c}$	Vectors composed by the level of inoperability and by the external failure, respectively.
$\mathcal{F}_P, \mathcal{F}_W$	Set of the most critical PNs and WNs, respectively.
$AD_{P,W}, AD_{W,P}$	Adjacency matrix representing the dependencies of the power system on the water system and the dependencies of the water system on the power system, respectively.
AD_P, AD_W	Adjacency matrix of the power and water systems, respectively.
$CI, C^{\text{level}}, C^{\text{loss}}$	Connectivity impact, level, and loss, respectively.
HL	Hydraulic head loss.
JD	Joint adjacency matrix of the power–water system.
PR	Joint failure probability matrix.
$\text{Set}_{PN}, \text{Set}_{WN}$	Set containing the top V PN and V WNs, respectively.
A	Dependence or influence matrix.
C	Hazen–Williams constant (unitless).
E_P, E_W	Set of power lines and pipes, respectively.
$P_{p,d}$	Probability mass function of node degree d .
V	Search space parameter.
V_P, V_W	Set of PNs and WNs, respectively.
<i>Variables</i>	
ϵ_i	Probability that PN i is no longer functional after a cascading failure event.
σ	Pipe diameter in inches.
τ	Flow rate in gallons per minute.
$\tilde{\tau}_p$	Normalized daily average flow of pipe p .
u_i	Probability that the generator connected to PN i is uncontrollable.
φ_i	Probability of breakage of pipe i .
ϖ_i	Dependence index of node i .
$\vec{R}_{P,i}, \vec{R}_{W,i}$	Vector of metrics for PN i and WN i , respectively.
ζ_j	Influence index of node j .
d_i	Degree of node i .
L	Pipe length in feet.
M_v	Weight of metric v obtained by AHP.

$P_{PN,i}, P_{WN,i}$	Probability of failure of PN i and WN i , respectively.
$R_{P,i}, R_{W,i}$	Overall vulnerability scores for PN i and WN i , respectively.
V_p	Vulnerability measure of pipe p .
$w_{d,t}$	Amount of water demand at time t .

I. INTRODUCTION AND MOTIVATION

URBAN critical infrastructures such as water and power grids play a vital role in the functions of present modern societies by enabling national public health and safety services among numerous other benefits. Pumping water through WDSs requires significant energy resources. The energy requirements of pumps differ due to the variations in the elevation and depth of WNs. The operation of water pumps is usually controlled via a time schedule dependent on the water demands, and/or a tank trigger level, where a pump is switched ON when the water level at the tank reaches a minimum level. During the electricity off-peak tariff periods, the pump is switched ON to keep the tank water level high enough to take advantage of the low electricity costs, while during peak tariff periods, the pump is only switched ON to maintain in the tank the minimum required water level. These considerations point out that for efficient energy usage, the two infrastructures need to operate jointly and interdependently [1]. This interdependence is further motivated by the reliance of water infrastructures on electricity from the power grid to operate, treat, store, and distribute water over a large area, and the power plants dependence on water in all stages of power generation process.

Cybersecurity represents a major challenge for power and water infrastructures, especially with the rise in the number of cyber-physical attacks in recent years. Such attacks can cause significant economic, social, and physical damage to the power-water infrastructures. In order to accurately conduct vulnerability assessment, one should not only rely on the topological information but also consider the hydraulic implications of the water system (pressure and flow variations, tank water level, demand patterns, etc.) as well as the power flow dynamics. Therefore, for a complete and accurate vulnerability assessment of the power-water system, we consider hydraulic, power, and topological information to propose a novel vulnerability assessment strategy, which we call the MCN-based vulnerability assessment. Such vulnerability assessment provides a clearer idea to operators on the technical exposure to cyber-physical attacks and the different vulnerabilities that an attacker can exploit to inflict a large damage on the system. Thus, the assessment strategy presented in this article would enable system defenders (operators) to focus their prioritization efforts on protecting the critical infrastructures against the damaging impact of attacks.

A. Related Work

Cascading failure analysis in interdependent power-water networks is understudied in the literature. In what follows, we start by highlighting the major works conducted in the interdependent power-water systems, then we summarize the works that target water systems solely and power systems solely. Finally, we list their common limitations.

1) *Works on the Interdependent Power-Water System:* Zhang et al. [2] studied the effect of the critical tolerance

parameter on the vulnerability of a hypothetical interdependent power-water system. Lee et al. [3] analyzed the vulnerability of an interdependent power-potable water supply system modeled as a Bayesian network by estimating the service disruption due to two types of failures, mainly common-cause failure events and cascading failure events. Najafi et al. [4] proposed modeling a framework of power-water systems with microgrids to minimize the investment cost of resilience strategies against hurricanes. Najafi et al. [5] studied the required resilience improvements in power-water systems in the presence of microgrids and natural disasters. Contingency analysis in coupled power-water systems was studied in [6], where Gorman identified the scenarios that violated the $N - 1$ contingency reliability. Khatavkar and Mays [7] proposed an analytical model to optimize in real-time the operation of water supply canal gates and pumps in the presence of limited power and/or water availability as well as natural and man-made disasters. Almoghathawi and Barker [8] quantified the effect of failure of the disrupted components on the overall resilience of interdependent power-water system, as well as the impact on resilience caused by the disruption of a single element. Pournaras et al. [9] simulated the impact of power cascading failures on the WDS to supply the required water demand to consumers. Wang et al. [10] proposed a framework to analyze two types of system vulnerabilities, namely, focused vulnerability and long-term vulnerability based on a redistribution flow model. Shuang et al. [11] identified critical nodes in the WDS in the presence of dynamic cascading failures and different tolerance parameters. Zhao et al. [12] proposed a two-stage mitigation strategy for water-energy systems against false data injection attacks (FDIAs). An inoculated subobserver-based fusion filter was proposed in [13] for power plants with desalination units that are prone to cyberattacks. The resilience of power systems under the interdependent operation of water systems was quantified in [14] for outages and drought conditions.

2) *Works on General Interdependent Infrastructures:* Another string of works deals with general physically interdependent infrastructures. Wang et al. [15] conducted a vulnerability analysis in critical infrastructures to quantify the importance of interdependent links against cascading failures, which were modeled based on a dynamic network flow redistribution model. Almoghathawi et al. [16] employed mixed integer programming to maximize the resilience of critical interdependent systems while minimizing at the same time the associated costs of the restoration under time and resource constraints. To maximize resilience of critical interdependent infrastructures, Ouyang [17] formulated the problem as a trilevel defender-attacker-defender model. An intrusion detection approach for SCADA systems controlling critical infrastructures was suggested in [18] using data-driven clustering of SCADA parameters.

The authors in [15], [16], [17], and [18] dealt with arbitrary physically interdependent infrastructures, and hence, they addressed mutual dependence from a different scope. Thus, these works constitute a research gap in the vulnerability assessment of power-water systems, especially that they did not identify the combination of target nodes that inflict the maximum damage on the system. In specific, the vulnerability assessment of [15] focused on evaluating the interdependence strength due to natural disruptive events, while the other works did not conduct a vulnerability assessment, rather they proposed a restoration, resilience, and intrusion detection models.

3) *Works on Water Systems:* Here, we summarize the works related to analyzing vulnerabilities in WDSs. Karamouz et al. [19] discussed the different vulnerability and risk assessment metrics of WDSs in face of natural hazards and threats. Albarakati et al. [20] compared different attack strategies and centrality measures to identify the most vulnerable components that have the largest damage impact on the water supply. In [21], to better understand the attack propagation and the response to attacks in WDSs, Adepu et al. proposed a design tool to launch single and multipoint stealthy attacks. Nicolaou et al. [22] proposed a strategy to reduce the vulnerability to cyber-physical attacks in WDSs by increasing the number of redundant cyber components in the graph describing the information flow from sensors to actuators.

4) *Works on Power Systems:* Finally, we briefly summarize the recent works on power systems vulnerabilities assessment. Jin et al. [23] solved the optimization problem whose objective is to find a stealthy FDIA vector that makes the ac-based state estimation spurious and misleading. Alexopoulos et al. [24] computed the minimal sets of measurements to be falsified for a successful FDIA. Pei et al. [25] first proposed a least-effort FDIA attack that targets the state estimation of power systems with minimal overhead in the presence of predeployed PMUs. Then, a PMUs predeployment greedy algorithm was proposed to protect the most vulnerable buses in the system. Jena et al. [26] proposed a coordinated cyber-physical attack strategy that maximizes the disruption of the electricity market operation by causing large deviations in the generation costs between the day-ahead market and real-time market operations. Liu et al. [27] proposed a coordinated sparse FDIA that required a minimal number of falsified measurements under full and incomplete topology information. Minimal sequential cyber topological attack mitigation strategy was analyzed in [28] using the concept of attack patterns that identifies the minimal attack sequence extracted from attack sequences. Malware attacks on power systems were proposed in [29], and different cyber protection strategies were compared in terms of their effectiveness in resisting malware attacks propagation in cyber-physical power systems.

5) *Common Limitations:* Despite the numerous contributions, the existing literature is limited by the following aspects.

- 1) Several works did not consider the hydraulic implications and behavior of the WDS in their vulnerability analysis, and based their analysis on hypothetical and probabilistic models (e.g. [2] and [3]), or even failed to assess the most vulnerable components that maximize the overall damage impact, such as [4], [5], [6], [7], [8], [10], [12], [13], and [14].
- 2) Multiple works assessed the system vulnerabilities from water system side solely, such as [11], [19], [20], [21], and [22]. In specific, these works considered hydraulic attributes and WDS-specific cyber-physical system in their vulnerability assessment, while ignoring the interdependent power components. Other works analyzed system vulnerabilities from power system side solely, such as [9], [23], [24], [25], [26], [27], [28], and [29]. In specific, Pournaras et al. [9] studied the effects of cascading failures in power systems on the water supply, while ignoring the mutual dependencies between both systems. As for the remaining works, they did not consider the dependent water system in their analysis.

- 3) None of the existing contributions addressed the problem of identifying the joint set of PNs/WNs that maximize the overall damaging impact on the system. This is important not only for designing robust systems against attacks but also for allowing government agencies to take proactive actions to protect the most vulnerable components in such interdependent systems.

B. Contributions and Organization

The contributions of this article are the following.

- 1) To the best of the authors' knowledge, none of the existing references identified the combination of PNs/WNs that cause the greatest damage on the system. This is important to design a robust system against attacks. Herein article, we identify such combination sets by proposing a low-complexity vulnerability assessment strategy.
- 2) We develop a joint CFS, which accounts for the power flow dynamics of the power system and the hydraulic behavior of the WDS. This allows to accurately quantify the impact of the cascading failure propagation and to assess the overall system damage.
- 3) We propose a novel vulnerability assessment strategy based on the MCN in the interdependent system. The proposed strategy uses a reduced search space of the joint set of PNs and WNs based on complex network theory, hydraulic, electrical, and topological features. The proposed strategy is compared to a set of benchmarks, and it is shown that it causes the maximum damage.
- 4) Our investigations reveal the following:
 - 1) the proposed MCN-based strategy identifies the most vulnerable node combinations that need to be protected in either or both the power and water systems, to avoid the highest damage impact on the system;
 - 2) the proposed strategy achieves a much lower computational time compared to the one that selects the target nodes based on their highest damage impact values, making it a promising approach for protecting critical infrastructures against targeted attacks and/or normal failures;
 - 3) the inflicting physical/hydraulic disruptions can be visualized under the identified most vulnerable combination of nodes.

The rest of this article is organized as follows. In Section II, the system model of the interdependent power–water network is presented. Section III infers the most critical set of PNs and WNs. Section IV describes the overall damage assessment and the joint CFS, while Section V presents the vulnerability assessment strategies. Experimental results and discussions are given in Section VI. Finally, Section VII concludes this article.

II. INTERDEPENDENT POWER–WATER MODEL

A. Power Grid Model

The power system is described as a graph $G_P = (V_P, E_P)$, where V_P is the set of PNs connected by power lines (edges) represented by E_P . Some of the PNs are generator nodes, while the rest represent load/demand nodes. PNs are characterized by unique voltage angles and magnitudes, while power lines are

characterized by active/reactive power flows determined by line impedance. To simulate the cascading failure in the power grid, we use MATLAB MATPOWER toolbox to run the power flow analysis using Newton's method, and MATCASC toolbox to model the resulting cascading failure. MATPOWER allows to determine the voltages, currents, and real and reactive power flows in the system, while MATCASC identifies the overloaded lines due to a target line removal. The power cascading failure disintegrates the system into islands, where power flow is redistributed among power lines, and lines overloading is checked again. An island is considered dead if it becomes free from a generation source.

B. Water Distribution System

The WDS consists of the following components:

- 1) reservoirs are sources of water such as rivers, groundwater, etc.;
- 2) tanks are nodes that store water whose levels can vary with time depending on actual demands;
- 3) pipes are links that transport water from one point to another;
- 4) pumps are devices that supply a constant amount of energy to a fluid so it can flow throughout the system;
- 5) valves are links that maintain or restrict the pressure or flow at a certain point;
- 6) junctions are nodes where water enters or leaves.

The WDS is a directed graph $G_W = (V_W, E_W)$, where V_W is the set of WNs (reservoirs, junctions, and tanks) characterized by a unique pressure head, and E_W is the set of links (pipes, valves, and pumps) characterized by a unique volumetric flow rate. The direction of each arc is dependent on the flow rate, which could be positive or negative. The pipes are subject to pressure head losses from the surface friction forces that act against the motion of the water, and they can be modeled using Hazen–Williams or Darcy–Weisbach formulas.

C. Interdependencies Modeling

The dependencies between power and water systems can be categorized as physical, geographical, cyber, and logical.

- 1) Physical dependencies such as water pumps and actionable valves requiring power to operate and hydropower/thermal power plants demanding water to generate electricity. Note that power plants that use solar, wind, diesel generators, and simple gas turbines do not require water for operation.
- 2) Geographical dependencies: the hydraulic and power components may be geographically colocated such as reverse-osmosis facilities located in the vicinity of power substations/plants; a storage tank depends on the nearest substation to deliver water to consumers, etc.

The joint adjacency matrix, JD, of the power–water system can be expressed as [30] $\begin{bmatrix} AD_P & AD_{P,W} \\ AD_{W,P} & AD_W \end{bmatrix}$ where AD_P and AD_W are the adjacency matrices of the power and water systems, respectively, $AD_{P,W}$ is the adjacency matrix representing the dependencies of the power system on the water system, and $AD_{W,P}$ is the adjacency matrix representing the dependencies of the water system on the power system. A joint probability matrix PR of same size as JD can be obtained by calculating the cascading failure probability of each component due

to the failure of an another component in a different infrastructure [31]. PNs can fail due to the following reasons: a direct or indirect nonfunctional generator substation and load shedding [32]. The load shedding occurs in the following two cases:

- 1) when the generator output becomes uncontrollable due to losing its connection to the control center, but the target PN still receives electricity supply from its neighbor nodes to supply its own load;
- 2) when the generator is no longer operating and all the neighboring nodes of the target PN have shed their load.

In the first case, the probability, ϕ_1 , that the load exceeds the uncontrollable generator output is [32]

$$\phi_1 = v_i \int_0^\infty \int_0^v f_L(u+v) f_G(u) du dv \quad (1)$$

where v_i is the probability that the generator connected directly or indirectly to PN i loses its communication path with the control center; f_L is the PDF of the load L of PN i , which can be modeled as an exponential distribution [33]: $f_L(u) = \tilde{L}^{-1} e^{-u/\tilde{L}}$, with $\tilde{L}$ being the mean of L ; and f_G is the PDF of the generator output G . In the second case, which accounts for a nonfunctional generator substation, PN i would need to shed its own load if it exceeds the generator's capacity $G_{\max}$. This probability can be determined as [32]

$$\phi_2 = \int_{G_{\max}}^\infty f_L(u) du = e^{-G/\tilde{L}}. \quad (2)$$

If $P_{p,d}$ is the probability mass function of node degree d , and ϵ_i is the probability that PN i is no longer functional after a cascading failure event, then $\sum_{d=0}^\infty P_{p,d} \epsilon_i^d$ is the probability that the neighbor nodes of the target PN i have shed their respective loads [32]. Finally, by the percolation theory, we can determine the average failure probability of PN i as

$$P_{PN,i} = \phi_1 + \phi_2 \sum_{d=0}^\infty f_d \epsilon_i^d. \quad (3)$$

Due to the existence of the infinite summation in (3), we make the following approximation as in [32] $\sum_{d=0}^\infty P_{p,d} \epsilon_i^d \approx P_{p,0} + P_{p,1} \epsilon_i$. On the other hand, the probability of failure of WNs depend on the following:

- 1) the probability of breakage of junctions/pipes;
- 2) the probability of the water demand exceeding the water supply.

In the first case, the failure probability of pipe i can be computed following the Poisson distribution as [34] $\varphi_i = 1 - e^{-\kappa_i}$, where $\kappa_i = r_i L_i$ is the expected number of failures per year for pipe i , r_i is the expected number of failures per year per unit length of pipe i , and L_i is the length of pipe i . In the second case, the probability of the water demand exceeding the water supply S can be computed as $\int_S^\infty f_W(u) du$, where $f_W(u)$ is the PDF of the water demand of the target WN, which can be modeled as a log logistic distribution using real water consumption data [35]. Finally, by the percolation theory, we can express the average failure probability of WN i as

$$P_{WN,i} = \varphi_i + \int_S^\infty f_W(u) du. \quad (4)$$

Then, we can compute the dependence or influence matrix, A , by multiplying the coefficients in JD by the coefficients in PR. The nonsymmetric matrix A with corresponding entry elements a_{ij} describes the extent to which the services degradation of

one infrastructure influences the services or operation of another infrastructure.

The interdependence between the power and water systems can be based on the concept of service degradation, which describes the extent to which the degradation in the operation and services offered by one infrastructure influences the availability of services and the level of operation in the other dependent infrastructures [36]. Based on this definition, we introduce the input–output inoperability model, where the inoperability is the inability of an infrastructure to perform its prescribed functions. The following input–output inoperability relationship can be defined as [36] $\mathbf{x} = \mathbf{A}\mathbf{x} + \mathbf{c}$, where $\mathbf{x}$ and $\mathbf{c}$ are vectors composed by the level of inoperability and by the external failure, respectively. While matrix $\mathbf{A}$ models the direct inoperability effects due to first-order dependencies, matrix $\mathbf{S} = \mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \dots$ models the inoperability effects due to first as well as higher order dependencies. Then, the overall inoperability due to a failure or perturbation $\mathbf{c}$ can be expressed as [36] $\mathbf{x} = \mathbf{S}\mathbf{c}$.

The inoperability level of the critical infrastructures is directly related to the failure of the most vulnerable components in the system, and that is because the failure of these nodes inflicts a large-scale cascading failure propagation that causes a large number of nodes to fail, leading thereby to a significant degradation in the quality of system services and operations. To quantify the robustness of one infrastructure due to the inoperability of the other infrastructures, Setola and De Porcellinis [37] introduced the dependence index $\varpi_i = \sum_j a_{ij}$ of node i , and the influence gain $\zeta_j = \sum_i a_{ij}$ of node j , which measures the influence that a node in one infrastructure has over the nodes in the other infrastructures.

III. CRITICALITY ASSESSMENT

This section focuses on inferring the set of the most critical PNs/WNs. It first introduces the different vulnerability metrics. Then, by employing the AHP, the corresponding weights are derived, which are later used in developing the MCN-based vulnerability assessment.

A. Metrics Definitions

For a system with the set of nodes $\mathcal{N} = \{1, 2, \dots, N\}$, with N denoting the total number of nodes, We define the following topological metrics that quantify the structural damage after a cascading failure event.

1) Topological Centrality Metrics:

- 1) Betweenness centrality indicates the extent to which a node lies on the shortest paths connecting other pairs of nodes. The shortest paths are based on breadth-first search algorithm, where the path length is measured by the number of edges. The betweenness centrality of node i is $BC_i = \sum_{j,p \neq i} \frac{n_{jp}(i)}{n_{jp}}$, where $n_{jp}(i)$ is the number of shortest paths between nodes j and p containing i .
- 2) Clustering coefficient indicates the extent to which nodes tend to form a cluster together: $CL_i = \frac{2\pi_i}{d_i(d_i-1)}$, where π_i is the number of connections between neighbors of node i and d_i is the degree of i .
- 3) Degree centrality is the number of nodes and edges that directly influence the status of a node: $CD_i = \frac{d_i}{N-1}$.
- 2) Structural Damage Metrics:

- 1) Connectivity impact (CI) indicates the number of nodes that remain connected after a node failure [38]. It is expressed as $CI = 1 - (N')/N$, where N' is the number of connected nodes after a node failure. This metric is applicable to both power and water distribution systems.
- 2) Connectivity level (C^{level}) indicates the average number of generators that are connected to load PNs [39]. This metric is specific to power systems solely.
- 3) Connectivity loss (C^{loss}) indicates the average decrease in the number of generators after a failure event [39]. It is expressed as $C^{\text{loss}} = (C^{\text{level}} - C^{\text{level,fail}})/C^{\text{level}}$, where $C^{\text{level,fail}}$ is the connectivity level after a failure event. This metric is specific to power systems solely.

The electrical centrality metrics and power flow distribution metrics are defined as follows.

3) Electrical Centrality Metrics:

- 1) Electrical betweenness centrality indicates the extent to which a node lies on the path connecting other pairs of nodes under the assumption that power flows over the shortest paths between them. It is expressed as $BC_i = \sum_{m \neq n \in \mathcal{N}} P_{mn}(i)/P_{mn}$, where $P_{mn}(i)$ is the power flow between PNs m and n flowing through PN i and P_{mn} denotes the power flow between PNs m and n .
- 2) Electrical degree centrality indicates the number of power flows that directly influence the status of node i . It is expressed as $CD_i = (\sum_{j \sim i} P_{ij})/(N-1)$, where $i \sim j$ means that node i is connected to node j and P_{ij} denotes the maximum electrical power flowing between i and j .

4) Power Flow Distribution Metrics:

- 1) Effective graph resistance captures the total cost to transfer power flow between a pair of nodes [40]. It is expressed as $R_G = \sum_{i=1}^N \sum_{j=i+1}^N R_{ij}$, where R_{ij} is the equivalent impedance between PNs i and j .
- 2) Load shedding measures the total apparent power after a node failure [38]. It is expressed as $LS = 1 - \sum_{i=1}^N (P_{D_i}^2 + Q_{D_i}^2)^{1/2} / \sum_{i=1}^N (P_{D_i}^2 + Q_{D_i}^2)^{1/2}$, where P_{D_i} and Q_{D_i} are the active and reactive power loads, respectively, that remain electrically connected after the failure of node i and P_{D_i} and Q_{D_i} are the active and reactive power loads before any node failure, respectively.
- 3) PoDN measures how well the power grid can supply power [41]. It is expressed as $PoDN = (E(G_P) - E(G'_P))/E(G_P)$, where $E(G_P)$ represents the net-ability before a failure event: $E(G_P) = \frac{1}{N_g \times N_l} \sum_{g \in \mathcal{G}} \sum_{l \in \mathcal{L}} P_{gl}/Z_{gl}$, where $\mathcal{G}$ is the set of generator nodes, $\mathcal{L}$ is the set of load nodes, N_g and N_l denote the number of generators and loads, respectively, P_{gl} stands for the capacity of power transmission between the pair of generation node g and load node l , and Z_{gl} represents the impedance between g and l [41].
- 5) Hydraulic Metrics: To identify critical WNs based on the hydraulic behavior, we use the following vulnerability measure [42]:

$$V_p(\alpha, \beta) = \alpha \tilde{\tau}_p + \beta \delta_{\text{next},p} + (1 - \alpha - \beta) N_{\text{prec},p}^{-1} \quad (5)$$

where α and β are weight coefficients selected as $\alpha = 1/5$ and $\beta = 1/9$ based on experiments in [42]; $\tilde{\tau}_p$ is the daily average flow of each pipe p normalized to the sum of all of the daily average flow in the network; $\delta_{\text{next},p}$ is the sum of the amount of

water conveyed by the successor pipes of p , normalized to the sum of water required to satisfy every consumer; and $N_{\text{prec},p}$ is the number of predecessor nodes from which a pipe p is reachable. In fact, if a pipe is reached from only a few nodes, it is considered vulnerable as it becomes more difficult to reach, hence, the -1 exponent.

B. Analytical Hierarchical Process (AHP)

With the help of different aforementioned metrics, we next aim to identify the weight impact of each metric on the overall system vulnerability in order to accurately obtain the failure criticality of each node. For this purpose, we turn toward AHP, which allows us to determine the most influential metrics that affect system vulnerability [43]. These weights are assigned based on the interactions between the metrics by performing pairwise comparisons in order to determine the relative importance of each metric with respect to the other. This means that each metric is assigned an impact value based on the extent to which it contributes to the overall system vulnerability. We first normalize all the metrics to the range of [0, 1] using min-max scaling to make them comparable. Then, we define a vector of metrics consisting of topological and electrical metrics for each PN i , $\vec{R}_{P,i}$, and a vector of metrics consisting of topological and hydraulic metrics for each WN i , $\vec{R}_{W,i}$. If for each node i , $\rho_{P,i}(b, v)$ and $\rho_{W,i}(b, v)$ are elements of $\vec{R}_{P,i}$ and $\vec{R}_{W,i}$, respectively, then the overall vulnerability scores for PN i and WN i can be expressed as [43]

$$R_{P,i} = \sum_{v=1}^{\eta_P} M_v \rho'_{P,i}(b, v); R_{W,i} = \sum_{v=1}^{\eta_W} M_v \rho'_{W,i}(b, v) \quad (6)$$

where η_P and η_W are the number of metrics for the power and water systems, respectively; M_v is obtained via AHP analysis, and it indicates the influence (weight) of metric v on the overall system vulnerability for a given system configuration; and $\rho'_{u,i}(b, v)$ is the linear transformation of $\rho_{u,i}(b, v)$, for $u = \{P, W\}$, given in [43, eq. (11)].

The most critical PNs/WNs are those nodes with vulnerability scores greater than certain thresholds ι_P and ι_W , respectively, i.e., the most critical PNs set is $\mathcal{F}_P = (V_P | R_{P,i} > \iota_P)$, and the most critical WNs set is $\mathcal{F}_W = (V_W | R_{W,i} > \iota_W)$. At the end of each cascading failure event, we re-calculate the weights of all of the metrics using AHP in order to reflect the changes in the new updated sets of $\mathcal{F}_P$ and $\mathcal{F}_W$.

IV. JOINT CFS

In this section, we introduce the joint CFS algorithms, which we will use in the next section to propose the MCN-based vulnerability assessment strategy.

A. Overall Damage Assessment

We start by quantifying the overall damage in the system after a failure event. The damage assessment of the WDS is evaluated using the DSR, which is defined as the ratio of the amount of water supplied, $w_{s,t}$, and the amount of water demanded, $w_{d,t}$ over a given time period t [44] $\text{DSR}_t = w_{s,t}/w_{d,t}$, with $0 \leq \text{DSR}_t \leq 1$, where 1 means that all water demands are supplied. The amount of water demand can be estimated as [44] $w_{d,t} = w_{d,1,t} + 0.015(\sum_p \tau_{p,t} - w_{d,1,t}) + 0.17\sqrt{\sum_p \tau_{p,t} - w_{d,1,t}}$, where $w_{d,1,t}$

Algorithm 1: Cascading Failure Simulator (CFS).

```

1: function CFS $\mathcal{P}, \mathcal{W}, \alpha_{\text{tol}}, \text{TN}$ : Dam
2: Input  $\mathcal{P}; \mathcal{W}; \alpha_{\text{tol}}$ : system tolerance; TN: target nodes list
3: Output Dam
4: for  $k = 1 \dots \text{length}(\text{TN})$  do
5:   if  $\text{TN}(k) \in \mathcal{P}$  then
6:      $[\mathcal{P}, \mathcal{W}, \text{Dam}] \leftarrow \text{CFS\_P}(\mathcal{P}, \mathcal{W}, \alpha, \text{TN}(k))$ 
7:   else
8:     if  $\text{TN}(k) \in \mathcal{W}$  then
9:        $[\mathcal{P}, \mathcal{W}, \text{Dam}] \leftarrow \text{CFS\_W}(\mathcal{P}, \mathcal{W}, \alpha, \text{TN}(k))$ 
10:    end if
11:  end if
12: end for

```

is the demand of the largest consumer and $\tau_{p,t}$ is the water flow at pipe p at time t .

The overall damage assessment is given as

$$\text{Dam} = \gamma_{P,1} \text{PoDN} + \gamma_{P,2} \text{CI}_P + \gamma_{W,1} (1 - \widetilde{\text{DSR}}) + \gamma_{W,2} \text{CI}_W \quad (7)$$

where $\gamma_{P,1}$ and $\gamma_{P,2}$ are the weight coefficients of the power grid; $\gamma_{W,1}$ and $\gamma_{W,2}$ are the weight coefficients of the WDS; $\widetilde{\text{DSR}}$ is the average DSR over all time periods; and CI_P and CI_W are the connectivity impact of the power grid and WDS, respectively. The weight coefficients are obtained via AHP analysis.

B. Cascading Failure Simulator (CFS)

Each of the power and water systems have their own dynamics. The power system can take under a second to reach equilibrium, while the water system can take minutes or hours to reach equilibrium. These dynamics differences are considered in the joint CFS, where the water system hydraulics span several hours to reach equilibrium before it starts impacting the power system.

Algorithm 1 sets up the cascading failure simulation for a joint PN-WN vulnerability assessment. Denote by $\mathcal{P}$ the power grid with N_P PNs, and by $\mathcal{W}$ the WDS with N_W WNs.

The algorithm starts by considering a target node TN, checking whether it is a PN or a WN, and then, calls the corresponding CFS. The function returns the overall damage Dam as well as the updated power grid $\mathcal{P}$ and WDS $\mathcal{W}$ topologies after the cascading failure.

Algorithm 2 simulates the cascading failure if the target node was a PN.

- 1) When the target PN fails, we first identify the set of PNs that become isolated from the rest of the network. We also search for those PNs that lost their paths to a generator substation (line 14).
- 2) Next, we identify the PNs that got overloaded from the cascading failure propagation (line 16), i.e., their corresponding extended betweenness¹ values after the cascading failure, EB_p^{new} are greater than the initial values, $\text{EB}_p^{\text{init}}$, multiplied by the system tolerance α_{tol} .
- 3) From the list of failing PNs, we identify those PNs that supply power to pumping stations and valves (line 20). Then, we simulate the cascading failure on these corresponding coupled pumps/valves (line 21).

¹The extended betweenness measure is the total transmission power drawn by a PN when the power is transmitted from generator PNs to load PNs.

Algorithm 2: Power Grid Cascading Failure.

```

1: CFS_PP,  $\mathcal{W}$ ,  $\alpha_{\text{tol}}$ , PN: Dam,  $\mathcal{P}$ ,  $\mathcal{W}$ 
2:  $\triangleright$  Failed WNs List (FWL): FWL  $\leftarrow \emptyset$ 
3:  $\text{EB}_{N_p}^{\text{init}} \leftarrow$  get initial extended betweenness for PNs
4:  $\triangleright$  Calculate capacity for PNs:  $K_{N_p} \leftarrow \alpha_{\text{tol}} \times \text{EB}_{N_p}^{\text{init}}$ 
5:  $\triangleright$  Failed PNs List (FPL): FPL  $\leftarrow \text{PN}$ 
6:  $\text{FPL}' \leftarrow \text{PN}$ 
7: while  $\text{FPL}' \neq \emptyset$  do
8:    $\text{FPL} \leftarrow \text{FPL} \cup \text{FPL}'$ 
9:   overloading  $\leftarrow 1$ 
10:  while overloading do
11:    overloading  $\leftarrow 0$ 
12:     $\text{EB}_{N_p}^{\text{new}} \leftarrow$  calculate new EB for PNs
13:    for  $i = 1 \dots N_p$  do
14:      if  $\text{isolated}(\text{PN}(i)) \parallel \sum_{j \neq \text{PN}(i)} \sum_{g \in G} e_{j,g}(\text{PN}(i)) = 0$ 
15:        then  $\text{FPL} \leftarrow \text{FPL} \cup \text{PN}(i)$ 
16:      end if
17:      if  $\text{EB}_{N_p}^{\text{new}}(i) > K_{N_p}(i)$  then overloading  $\leftarrow 1$ ,
18:         $\text{FPL}' \leftarrow \text{FPL}' \cup \text{PN}(i)$ 
19:      end if
20:    end for
21:  end while
22:   $\text{FPL}'' \leftarrow$  set of all PNs that feed pumps/valves
23:  if  $\text{FPL}'' \neq \{\}$  then simulate failure on the target
24:    pumps/valves,  $\text{FWL}' \leftarrow$  failed WNs,  $\text{FWL} \leftarrow \text{FWL} \cup \text{FWL}'$ 
25:  end if
26:   $\text{FWL}'' \leftarrow$  set of all WNs  $\in \text{FWL}'$  with geo-physical
27:    dependence with newly failed PNs  $\in \text{FPL}$ 
28:   $\text{FPL}' \leftarrow$  corresponding PNs of WNs  $\in \text{FWL}''$ 
29: end while
30:  $\mathcal{W} \leftarrow$  switch off all failed WNs in FWL
31:  $\mathcal{P} \leftarrow$  remove all failed PNs in FPL
32: Measure the damage using (7)

```

- 4) Next, we identify the failing WNs that have geographical/physical dependencies with PNs (line 23), and these PNs get added to the failing list of PNs (line 24).
- 5) The cascading failure keeps propagating back and forth between the power system and WDS until there are no more new failed PNs.
- 6) Finally, we update the WDS and power grid topologies (lines 26 and 27), and we assess the overall damage in the system using (7).
- 7) Similar to Algorithm 2, we simulate the cascading failure if the initial target node is a WN.

It is worthy to note that WNs fail if [45]

- 1) the pressure at the corresponding node is negative;
- 2) the pressure falls below the minimum pressure causing flow reductions or blanking;
- 3) the pressure is greater than the maximum pressure, which can cause pipe leakage and loss of service function.

In the first two conditions, the system cannot supply sufficient pressure to satisfy the given water demand, while in the last condition, the corresponding pipe can no longer withstand the pressure of the fluid, which eventually breaks it.

The maximum pressure is proportional to the operating pressure. The operating pressure is the summation of static head and dynamic head, where the static head is the elevation change between the surface of the reservoir and the point of discharge in

the receiving water tank, while the dynamic head is the height at which the fluid needs to be pumped considering friction losses in the pipe. The dynamic head is calculated as the difference between static head and head loss, where the head loss, HL, is the loss of energy of the fluid due to friction, length, fittings, and pipes bends expressed as $\text{HL} = 10.67L\tau^{1.852}/C^{1.852}\sigma^{4.8704}$, where L is the pipe length in meters; τ is the flow rate in m^3/s ; C is the Hazen–Williams roughness coefficient; and σ is the pipe diameter in meters.

V. JOINT VULNERABILITY ANALYSIS

In this section, we describe different vulnerability assessment strategies for selecting the best combination of PNs/WNs that present the greatest damage on both subsystems. For all the following strategies, we obtain the overall vulnerability strength using (7).

A. Proposed MCN Joint PNs–WNs Strategy

The objective of the joint MCN strategy is to obtain m -nodes combinations of PNs and WNs that maximize the overall damage impact. To build up to sets of m -nodes combinations, we start with a group of nodes consisting of $m = 1$ node. There are a total of $\binom{N_p + N_w}{m}$ single node combinations to choose from. However, we would like to choose the most vulnerable PNs/WNs. For this purpose, we obtain Set_{PN} containing the top V PNs that are the most critical from the set $\mathcal{F}_p$, and Set_{WN} containing the top V WNs that are the most critical from the set $\mathcal{F}_w$. Note that the sets $\mathcal{F}_p$ and $\mathcal{F}_w$ are obtained via AHP analysis as described in Section III-B. The parameter V can be selected as a small value between 4 and 8 to limit the search space [41]. If $V = 4$, then the first set, made up of $m = 1$ target node, would consist of the top four nodes from Set_{PN} and the top four nodes from Set_{WN} .

To obtain set 2 with $m = 2$ -nodes combinations, we first construct all the possible 2-node combinations by combining each node from Set_{PN} and Set_{WN} with the nodes from the previous set, i.e., set 1. Then, we search over all these possible combinations, the ones that inflict the maximum damage on the system. Set 2 can include three potential groups: the first group consisting of two PNs, the second group consisting of two WNs, and the third group consisting of a single PN and a single WN. Note that not all of these groups need to be present in the final selected most vulnerable combinations. In this case, set 2 would consist of the twelve most damaging 2-node combinations. To generalize on building the subsequent m th set of m -node combinations, we construct all the possible m -node combinations by combining the node combinations from the previous $(m - 1)$ th set with each node from Set_{PN} and Set_{WN} . The m th set can include $(m + 1)$ groups, where each group includes V combinations of the most critical PNs and WNs. Then, we select from all these possible combinations, the $(m + 1)V$ combinations that maximize the overall damage.

Remarks: AHP is an objective decision-making strategy that minimizes ambiguity and boosts confidence in results by minimizing bias. However, the weights assignment can be subjective, as they change depending on the system under study. Moreover, a larger starting set of critical nodes leads to a larger search space and increased computational time. In summary, the proposed vulnerability assessment strategy is a general objective approach that apply to any interdependent system of any size. However, the subjectivity of the methodology (the weight values and the



Fig. 1. Graphical representation of C-Town WDS [47].

search space size) only arises from its actual application to the specific system under study.

Computational complexity: Let M denote the maximum number of joint PNs/WNs to target, then the theoretical complexity of the proposed MCN strategy is $(V^2 \times M) \times \mathcal{O}_{\text{CFS}}$, where $\mathcal{O}_{\text{CFS}}$ is the computational complexity of the CFS, shown in [46] to be hard to quantify.

B. Benchmark Strategies

1) **MCN-PNs Strategy:** This strategy consists of target PNs only. The first set of a single target PN consists of the top $2V$ most critical PNs based on the set $\mathcal{F}_P$. The subsequent m th set of node combinations is obtained by combining the previous set with each node from the top $2V$ most critical PNs. All of the possible node combinations are sorted in descending order according to their corresponding damage values, and finally the top $(m+1)V$ combination sets are selected.

2) **MCN-WNs Strategy:** This strategy consists of target WNs only. The first set of a single WN consists of the top $2V$ most critical WNs based on $\mathcal{F}_W$. The subsequent m th sets are obtained by combining the previous set with each node from the top $2V$ most critical WNs. All of the possible node combinations are sorted in descending order based on their damage values, and the top $(m+1)V$ sets are selected.

3) **Damage-Based Strategy:** This strategy evaluates the overall damage impact of each PN and WN in the system. Then, PNs and WNs are sorted in descending order based on their damage impact values, and the top $2V$ most damaging PNs/WNs are selected. The subsequent m th set of node combinations is obtained by combining the previous set with each node from

the top $2V$ most damaging PNs/WNs. All of the possible node combinations are sorted in a descending order according to their corresponding damage values, and finally, the top $(m+1)V$ combination sets are selected.

VI. EXPERIMENTAL ANALYSIS

A. Computational Environment

The system configuration is a 64-bit Microsoft Windows 10 Home with an Intel Core i9-10980HK CPU at 2.4 GHz, along with an installed physical memory (RAM) of 32 GB. We use IEEE 118-bus system that consists of 118 buses and 186 links. We use the benchmark model C-Town based on a real-world medium-sized system consisting of 429 pipes, 396 nodes, among which 388 are demand nodes, one reservoir, seven tanks (T1–T7), 11 pumps (PUs) constituting five pumping stations S1–S5, and four valves, among which one is actionable valve, V1, that controls the water level in tank T2. Fig. 1 shows a graphical representation of C-Town WDS. Moreover, we use the 39-bus New England system coupled with Richmond Skeleton WDS, which consists of consists of 48 nodes, among which seven are pumps, six are storage tanks, and one is a reservoir. The physical dependencies of the C-Town and 118-bus are as follows [9]: pumping station S1 is coupled with PN 108, S2 is coupled with PN 118, S3 is coupled with PN 70, S4 is coupled with PN 17, S5 is coupled with PN 20, and actionable valve V1 is coupled with PN 83. As for the geographical dependencies, they are obtained by loosely overlaying C-Town WDS with IEEE 118-bus system, as provided by Table I. As for the physical dependencies of Richmond WDS and 39-bus, they are as follows: bus 23 with

TABLE I
GEOGRAPHICAL DEPENDENCIES OF C-TOWN WDS AND
IEEE 118-BUS SYSTEM

WN	PN	WN	PN
280	108,109	301	20,35,38
285	104,105,106,107	302	36
276	103,110,111,112	304	30
273	100	306	18
274	101,102	220	17
156	94	86	19
154	91,92,93,95	87	34,37
155	96	85	33
160	88,89,90	237	12,13,14
159	83,84,85	57	16
162	86,87	62	6,7,8
1058	82	76	4
213	98,99	54	3
216	79,97,80	64	1,2
197	81	358	25
227	78	352	23
258	76	1208	22
238	75,118	382	24
419	70,74	360	26
422	71	320	27
231	72	330	28
288	68,69	297	29,31
290	65	329	32
291	66	267	115
19	67	354	114
425	47	305	19
426	48	323	21
363	50,51	332	113
93	57,58	1153	9
233	45	118	5
234	43	305	10
385	44	67	11
134	39,40,41	242	15
251	42	307	46
257	52	441	49
30	56	90	53,54
36	59	27	55
88	63	317	73
28	60	258	77
81	61,64	238	116
350	62	65	117

PU 1 and PU 2; bus 19 with PU 3; bus 15 with PU 4; bus 6 with PU 5; bus 11 with PU 6; and bus 8 with PU 7. Similarly, we obtain the geographical dependencies by overlaying the WDS with the power system.

The network maps of C-Town and Richmond Skeleton WDSs are obtained using a public water modeling software program called EPANET developed by the U.S. Environmental Protection Agency's Water Supply and Water Resources Division. The WDS graph trees are obtained using Water Network Tool for Resilience, an EPANET Python library. The network map and graph tree are then input into MATLAB, where all hydraulic and power simulations take place. In specific, we use the following MATLAB toolboxes:

- 1) MATPOWER to run ac power flow;
- 2) MATCASC to model the power cascading failures;
- 3) epanetCPA [48] to simulate WDS failures and assess the hydraulic response over 24 h.

The WDS failures can be due to malfunctioning, faults, or cyber-physical attacks. Compared to the power dynamics, this takes into consideration the slow hydraulic dynamics to reach equilibrium. The toolbox extends EPANET features to include a cyber layer of sensors, actuators, PLCs, and a SCADA system. This allows to model failures on tanks, pumps, and valves by

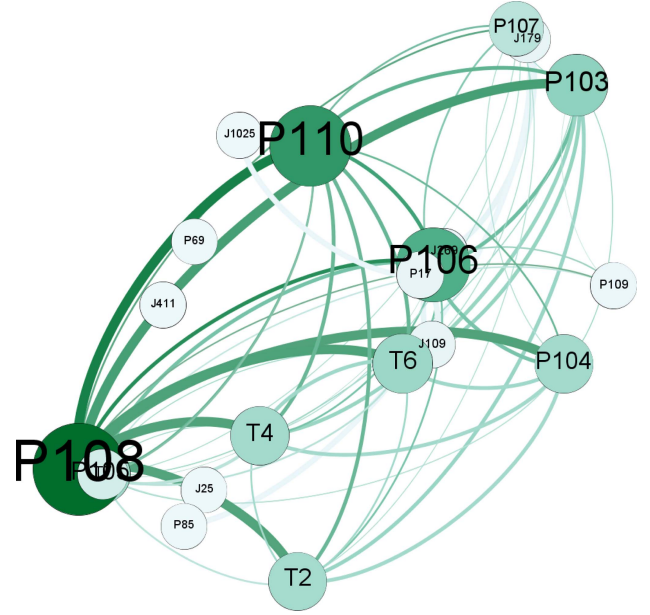


TABLE II
MOST DEPENDENT PNs/WNS BASED ON THE CALCULATED DEPENDENCE INDEX FOR 118-BUS COUPLED WITH C-TOWN WDS

Node	P75	P85	P49	P12	J301	P100	J276	J285	P15	P105
Dependence index %	100	97.19	97.15	93.38	92.5	91.99	86.35	86.22	85.76	85.55

TABLE III
MOST INFLUENTIAL NODES BASED ON THE CALCULATED INFLUENCE INDEX FOR 118-BUS SYSTEM COUPLED WITH C-TOWN WDS

Node	J301	J285	J276	J237	J159	J273	J154	J304	J216	J291
Influence index %	100	99.23	87.31	72.81	72.57	67.94	65.19	63.99	61.39	59.71

TABLE IV
COMBINATION SETS OF PNs/WNS FOR THE MCN-BASED STRATEGY FOR 118-BUS SYSTEM COUPLED WITH C-TOWN WDS

Index	Set 1	% Damage	Set 2	% Damage	Set 3	% Damage	Set 4	% Damage
1	P108	80.22	P108 P110	88.46	P108 P110 P103	88.68	P108 P110 P106 P105	89.1
2	P70	48.79	P108 P104	87.83	P108 P110 P106	88.68	P108 P110 P103 P106	88.89
3	P69	42.89	P108 P103	87.43	P108 P110 T6	88.53	P108 P110 P106 P104	88.89
4	P17	34.71	P108 T4	86.37	P108 P110 T2	88.53	P108 P110 P106 P107	88.89
5	J25	74.26	P108 P105	81.8	P108 P110 T4	88.53	P108 P110 P103 T6	88.74
6	J179	74.04	P108 T6	80.38	P108 P104 T6	87.89	P108 P110 P103 T2	88.74
7	J109	72.86	P108 T2	80.36	P108 P104 T2	87.89	P108 P110 P103 T4	88.74
8	J411	34.83	P17 J109	77.35	P108 P104 T4	87.89	P108 P110 P106 T6	88.74
9			J179 J269	77.25	P108 P103 P106	87.64	P108 P110 P106 T2	88.74
10			J109 J269	77.25	P108 P103 T6	87.5	P108 P110 P106 T4	88.74
11			P17 J1025	77.24	P108 P103 T2	87.5	P108 P110 T6 T4	88.59
12			J179 P85	76.01	P108 P103 T4	87.5	P108 P110 T6 T2	88.59
13					P108 P105 P106	87.27	P108 P110 T6 T7	88.59
14					P108 P104 P106	87.01	P108 P110 T2 T4	88.59
15					P108 T4 T7	86.96	P108 P110 T2 T7	88.59
16					P108 T4 T6	86.44	P108 P110 T4 T7	88.59
17							P108 P105 P106 P109	88.46
18							P108 P104 T6 T4	87.96
19							P108 P104 T6 T2	87.96
20							P108 P104 T6 T7	87.96

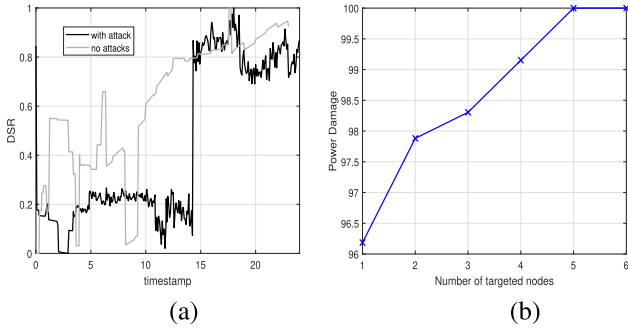


Fig. 3. (a) Normalized DSR versus timestamp duration. (b) Power damage versus the number of target nodes for C-Town WDS.

and Set 4 consist of 8, 12, 16, and 20 joint PNs/WNs, respectively, sorted according to their damage impact values. One of the combination sets that causes the largest damage impact is {P108, P110, P106, P105, P103, T6}.

Based on the combination set with the largest damage impact, we visualize in Fig. 2 the resulting risk graph. The bigger and darker the vertices and the links are, the more critical they are in terms of contributing to the overall damage. We can see that “P106,” “P108,” and “P110” are the nodes with the greatest power damage impact, while “T2,” “T4,” and “T6” are the nodes with the greatest water damage impact.

Fig. 3(a) shows the normalized DSR values under normal and failure (based on the combination set {P108, P110, P106, P105, P103, T6}) conditions for the

C-Town WDS. The DSR values were calculated at each time step in the simulations based on the supply under failure and the demand under normal operation. In this context, DSR measures the amount of demand water (under normal operation) we can satisfy when the supply is under failure. Through the simulations, we found out that when the system is targeted using nodes {P108, P110, P106, P105, P103, T6}, tanks T1–T6 and pumps 1–11 fail. This explains why the ratio of water supply to demand becomes 0 up to hour 15 as the cascading failure propagates, since the failure of tanks and pumps led to customers’ full water demand not being met, i.e., there was a deficit in the supplied water quantity. However, the DSR goes up half-way due to the water demand under normal operations being low during this period. Moreover, Fig. 3(b) shows the power damage values in terms of weighted PoDN and CI for set {P108, P110, P106, P105, P103, T6}. As can be seen, the damage values increase up to 100% with five target nodes.

To illustrate the advantages of the proposed MCN strategy, we compare it in Fig. 4 to the MCN–PNs vulnerability assessment strategy, the MCN–WNS strategy, and the damage-based strategy for both C-Town and Richmond Skeleton WDSs. Comparing the proposed MCN joint PNs–WNS strategy with the damage-based strategy, we see from Fig. 4 that the damage-based strategy is slightly higher than the MCN PNs–WNS strategy; however, the computational time of the damage-based strategy is much higher than the proposed strategy, as shown in Fig. 5. In specific, compared to the proposed MCN joint PNs–WNS, the computational time of the damage-based strategy was on average 108.4% and 278.96% higher in obtaining a set of two

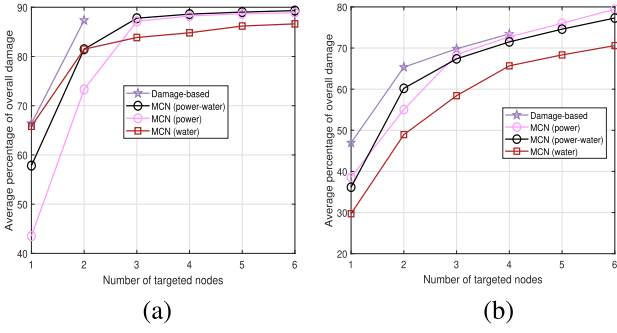


Fig. 4. Overall damage versus the number of target PNs/WNs ($V = 4$) for (a) C-Town and (b) Richmond WDS.

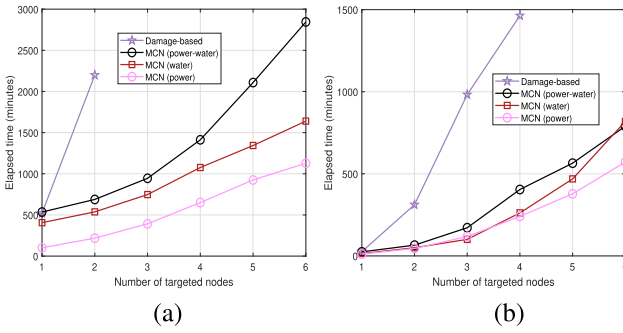


Fig. 5. Time complexity of different vulnerability assessment strategies for (a) C-Town and (b) Richmond WDS.

and four target nodes for the C-Town and Richmond WDSs, respectively. This translates to around double and triple increase in simulation time relative to the proposed strategy for the C-Town and Richmond WDSs, respectively. This indicates that the MCN-based strategy can identify the node combinations that cause a high damage impact on the system but with a much less computational time. Moreover, a comparable performance to the MCN joint PNs–WNs strategy is observed with MCN–PNs strategy for both WDSs. Fig. 6 shows a comparison between pumps/valve flows under normal and failure conditions based on the combination set $\{P108, P110, P106, P105, P103, T6\}$ for the C-Town WDS. We can see that the flows at all the pumps and valve V1 drop to 0 after the targeted failure.

Finally, based on the MCN strategy with the set $\{P108, P110, P106, P105, P103, T6\}$ for the C-Town WDS, we plot in Fig. 7, the hydraulic pressures of the six tanks, and we compare them to when the system is normally operating. We can see that the tank pressures go to 0 just few hours after the attack has completely cascaded.

VII. CONCLUSION

In this article, we developed a joint CFS for power–water system by identifying the MCNs based on AHP analysis. Then, based on the obtained MCNs, we proposed a low-complexity vulnerability assessment strategy to identify the joint set of vulnerable nodes. We showed that this strategy identified the most vulnerable node combinations that cause a high damage impact in the system but with a much lower computational time compared to the strategy that selects the vulnerable nodes

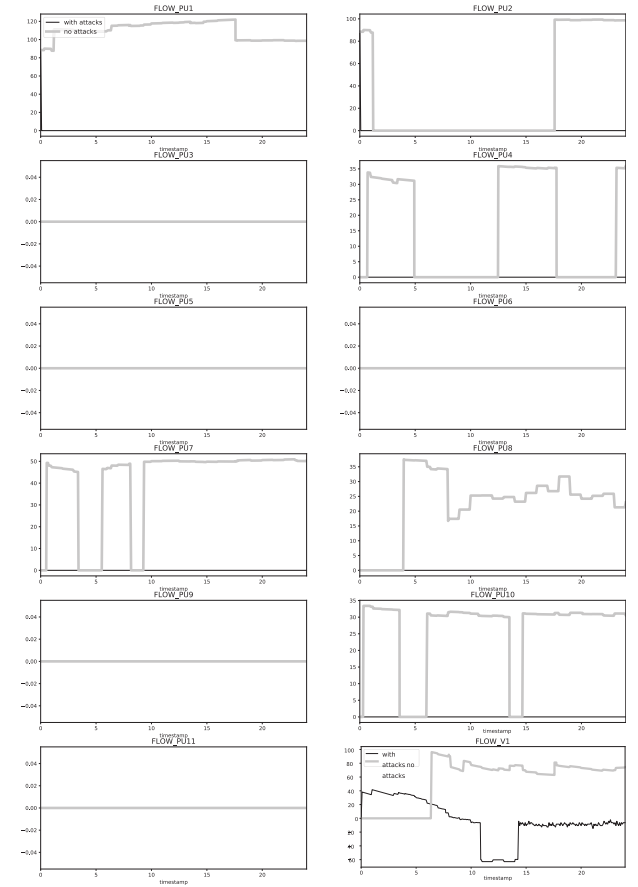


Fig. 6. Comparison between pumps/valve flows of C-Town over a 24-h period with a timestep of 0.0833 min under normal and targeted failure conditions.

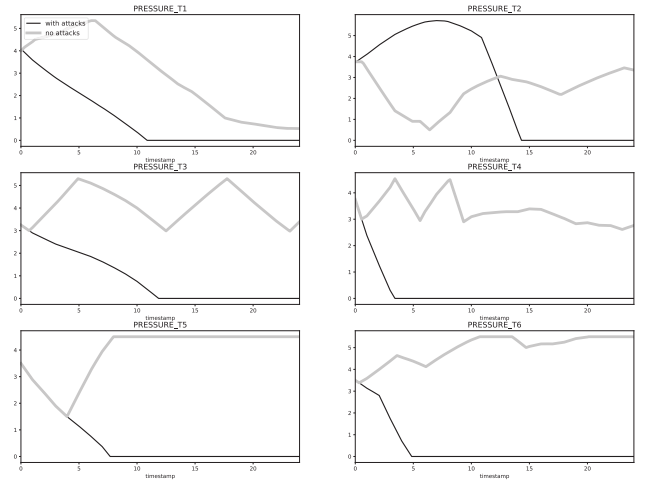


Fig. 7. Comparison between tank pressures of C-Town over a 24-h period with a timestep of 0.0833 min under normal and targeted failure conditions.

based on their highest damage values. Our findings have important implications for designing a robust power–water infrastructure and for minimizing the impact of disruptions.

ACKNOWLEDGMENT

The statements made herein are solely the responsibility of the authors.

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